

Analysis First-Year Exam, August 2021, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be continuous. Prove that if $\limsup_{n \rightarrow \infty} f(n) = 1$ and $\liminf_{n \rightarrow \infty} f(n) = -1$ then f has infinitely many zeros.
2. Let $f : X \rightarrow Y$ be a continuous bijection between metric spaces. For each of the following statements, give a proof or a counterexample, as appropriate:
 - (a) If X is compact then f^{-1} is continuous;
 - (b) if Y is compact then f^{-1} is continuous.
3. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = 0$. Prove that if f is continuous then it is uniformly continuous.
4. Let $(a_n)_{n=0}^{\infty}$ be a sequence of positive reals. Prove that if the series $\sum_{n=0}^{\infty} a_n$ converges then so does the series $\sum_{n=0}^{\infty} \sqrt{a_n a_{n+1}}$. Prove also that the converse holds when $(a_n)_{n=0}^{\infty}$ is decreasing.
5. Let the function f be defined on an open interval about 0; assume that $f(0) = 0$ and that f is differentiable at 0. When n is a positive integer, determine the value of the limit

$$\lim_{x \rightarrow 0} \frac{1}{x} \left[f(x) + f\left(\frac{x}{2}\right) + f\left(\frac{x}{3}\right) + \cdots + f\left(\frac{x}{n}\right) \right].$$